Worksheet 6.5 — The Cross Product

Grade 12 Calculus and Vectors (MCV4U) · Unit 6: Geometry and Algebra of Vectors

The Big Ideas

$\vec{a} \times \vec{b} = (a_2b_3 - a_3b_2,\ a_3b_1 - a_1b_3,\ a_1b_2 - a_2b_1)$ is perpendicular to both; $|\vec{a} \times \vec{b}|$ is the parallelogram area.

- $\vec{a} \times \vec{b} = (a_2b_3 - a_3b_2,\ a_3b_1 - a_1b_3,\ a_1b_2 - a_2b_1)$.
- $\vec{a} \times \vec{b} \perp \vec{a}$ and $\perp \vec{b}$.
- $|\vec{a} \times \vec{b}|$ = parallelogram area; triangle $= \frac{1}{2}|\vec{a} \times \vec{b}|$.

The Big Picture

Multiplying to a vector

The **cross product** $\vec{u} \times \vec{v}$ produces a *vector* perpendicular to both $\vec{u}$ and $\vec{v}$ in 3-D.

Magnitude and area

Its magnitude $|\vec{u}|\,|\vec{v}|\sin\theta$ equals the area of the parallelogram the two vectors span.

Direction

The right-hand rule fixes the direction, and $\vec{u} \times \vec{v} = -\vec{v} \times \vec{u}$ order matters.

Worked Examples

Example 1 — Standard basis

Compute $(1,0,0) \times (0,1,0)$.

Solution. **Step 1 (apply the cross-product formula $(a_2b_3 - a_3b_2,\ a_3b_1 - a_1b_3,\ a_1b_2 - a_2b_1)$ with $\vec{a} = (1,0,0), \vec{b} = (0,1,0)$):** first component $0 \cdot 0 - 0 \cdot 1 = 0$; second $0 \cdot 0 - 1 \cdot 0 = 0$; third $1 \cdot 1 - 0 \cdot 0 = 1$.

Conclusion: $(1,0,0) \times (0,1,0) = (0,0,1)$.

Example 2 — General

Compute $(1,2,3) \times (4,5,6)$.

Solution. **Step 1 (label components):** $\vec{a} = (1,2,3)$, $\vec{b} = (4,5,6)$.

Step 2 (apply the formula component by component): first: $2 \cdot 6 - 3 \cdot 5 = 12 - 15 = -3$; second: $3 \cdot 4 - 1 \cdot 6 = 12 - 6 = 6$; third: $1 \cdot 5 - 2 \cdot 4 = 5 - 8 = -3$.

Conclusion: $(1,2,3) \times (4,5,6) = (-3,6,-3)$.

Example 3 — Perpendicular check

Verify $(-3,6,-3) \perp (1,2,3)$.

Solution. **Step 1 (use the dot product to test perpendicularity):** $(-3)(1) + (6)(2) + (-3)(3)$.

Step 2 (compute): $-3 + 12 - 9$.

Conclusion: $= 0$, so the vectors are perpendicular (as the cross product must be, to both original vectors).

Example 4 — Parallelogram area

Area from $(1,0,0)$ and $(0,2,0)$.

Solution. **Step 1 (compute the cross product):** $(1,0,0)\times(0,2,0) = (0\cdot0-0\cdot2,\ 0\cdot0-1\cdot0,\ 1\cdot2-0\cdot0) = (0,0,2)$.

Step 2 (the parallelogram area equals the magnitude of the cross product): $|(0,0,2)| = 2$.

Conclusion: area $= 2$.

Example 5 — Triangle area

Triangle area from $(1,0,0)$ and $(0,2,0)$.

Solution. **Step 1 (recall the triangle is half the parallelogram):** from the previous example, the parallelogram area is 2.

Step 2 (halve it): $\frac{1}{2}(2)$.

Conclusion: triangle area $= 1$.

Example 6 — Standard basis

Compute $(0,1,0)\times(0,0,1)$.

Solution. **Step 1 (apply the formula with $\vec{a} = (0,1,0), \vec{b} = (0,0,1)$):** first: $1\cdot1-0\cdot0=1$; second: $0\cdot0-0\cdot1=0$; third: $0\cdot0-1\cdot0=0$.

Conclusion: $(0,1,0)\times(0,0,1) = (1,0,0)$.

Example 7 — General

Compute $(2,0,0)\times(0,3,0)$.

Solution. **Step 1 (apply the formula):** first: $0\cdot0-0\cdot3=0$; second: $0\cdot0-2\cdot0=0$; third: $2\cdot3-0\cdot0=6$.

Conclusion: $(2,0,0)\times(0,3,0) = (0,0,6)$.

Example 8 — General

Compute $(1,1,0)\times(0,1,1)$.

Solution. **Step 1 (apply the formula with $\vec{a} = (1,1,0), \vec{b} = (0,1,1)$):** first: $1\cdot1-0\cdot1=1$; second: $0\cdot0-1\cdot1=-1$; third: $1\cdot1-1\cdot0=1$.

Conclusion: $(1,1,0)\times(0,1,1) = (1,-1,1)$.

Example 9 — Parallelogram area

Area from $(3,0,0)$ and $(0,4,0)$.

Solution. **Step 1 (compute the cross product):** $(3,0,0)\times(0,4,0) = (0\cdot0-0\cdot4,\ 0\cdot0-3\cdot0,\ 3\cdot4-0\cdot0) = (0,0,12)$.

Step 2 (take the magnitude): $|(0,0,12)| = 12$.

Conclusion: area $= 12$.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Compute $(0, 1, 0) \times (0, 0, 1)$.

Question 2

Compute $(2, 0, 0) \times (0, 3, 0)$.

Question 3

What direction is $\vec{a} \times \vec{b}$ relative to $\vec{a}, \vec{b}$?

Question 4

Area of the parallelogram from $(3, 0, 0)$ and $(0, 4, 0)$?

Question 5

Compute $(1, 1, 0) \times (0, 1, 1)$.

Question 6

Compute $(1, 0, 0) \times (0, 0, 1)$.

Question 7

Compute $(1, 2, 3) \times (4, 5, 6)$.

Question 8

Triangle area from $(2, 0, 0)$ and $(0, 2, 0)$?

Question 9

Is $(1, 2, 3) \times (2, 4, 6)$ the zero vector? Why?

Question 10

Compute $(0, 0, 1) \times (1, 0, 0)$.

Question 11

Magnitude of $(1, 0, 0) \times (0, 5, 0)$?

Question 12

Verify $(1, 2, 3) \times (4, 5, 6) = (-3, 6, -3)$ is perpendicular to $(4, 5, 6)$.

Question 13 — Challenge

Find the area of the triangle with vertices $A(0, 0, 0)$, $B(1, 0, 0)$, $C(0, 2, 3)$.

Question Recap & Answer Key

Question Recap

1. Compute $(0,1,0) \times (0,0,1)$.
2. Compute $(2,0,0) \times (0,3,0)$.
3. What direction is $\vec{a} \times \vec{b}$ relative to $\vec{a}, \vec{b}$?
4. Area of the parallelogram from $(3,0,0)$ and $(0,4,0)$?
5. Compute $(1,1,0) \times (0,1,1)$.
6. Compute $(1,0,0) \times (0,0,1)$.
7. Compute $(1,2,3) \times (4,5,6)$.
8. Triangle area from $(2,0,0)$ and $(0,2,0)$?
9. Is $(1,2,3) \times (2,4,6)$ the zero vector? Why?
10. Compute $(0,0,1) \times (1,0,0)$.
11. Magnitude of $(1,0,0) \times (0,5,0)$?
12. Verify $(1,2,3) \times (4,5,6) = (-3,6,-3)$ is perpendicular to $(4,5,6)$.
13. Find the area of the triangle with vertices $A(0,0,0)$, $B(1,0,0)$, $C(0,2,3)$.

✓ Answer Key

1. $(1,0,0)$
2. $(0,0,6)$
3. perpendicular to both
4. 12
5. $(1,-1,1)$
6. $(0,-1,0)$
7. $(-3,6,-3)$
8. $\frac{1}{2}(4) = 2$
9. yes they are parallel ($\sin\theta = 0$)
10. $(0,1,0)$
11. 5
12. $-12 + 30 - 18 = 0$
13. $\vec{AB} \times \vec{AC} = (1,0,0) \times (0,2,3) = (0,-3,2)$; area $= \frac{1}{2}\sqrt{13}$